



Several Recent Results Related to This Conference:
**Dust Coagulation in VSI, Eccentric Mode Instability in
Self-Gravitating PPDs, and Global Dust Flows**

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Many collaborators over the years

LA-UR-26-21208



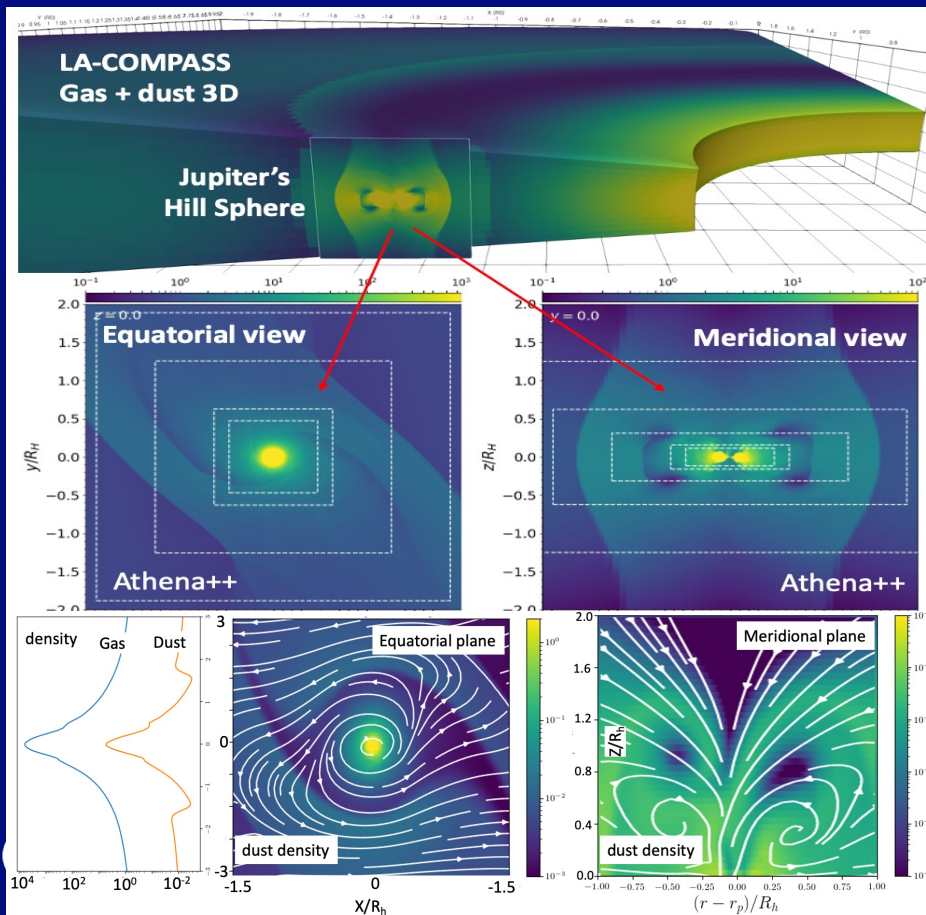
Outline

- Dust modeling in multi-fluid PPDs (**LANL+LMU coll.**)
- **A)** VSI w. dust Coagulation, Fragmentation and Bouncing (**K. Ho, HL, S. Li 2026, to be submitted**)
- **B)** Eccentric Mode Instability (EMI) in Self-Gravitating Disk (**Jiaru Li et al. 2021**)
- **C)** Global dust supply (**H. Li and S. Li 2026, in preparation**)
- Summary and outlook



2D/3D Two-Fluid (gas+dust) Simulations of PPDs w (multiple) Planets

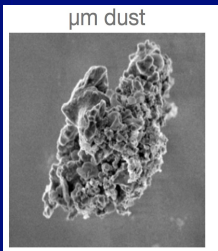
- Jupiter mass planet
- 3D gas + several dust sizes
- 2D (r - ϕ , r - θ) w dust size evol.
- ~ 3500 rotations (0.57 Myr at 30 au)



HL & S. Li 2026, in prep

Multiscale/ Multi-physics/ Multi-species/ Multiphase theory and modeling

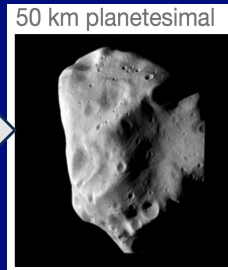
Dust
growth



$\mu\text{m} - \text{cm}$

$10^{-19/-13} R$

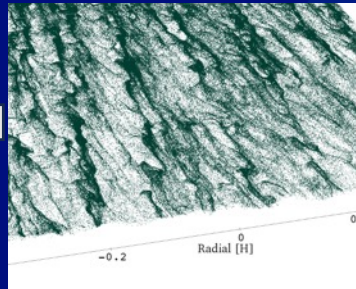
Planetesimals



$10-100 \text{ km}$

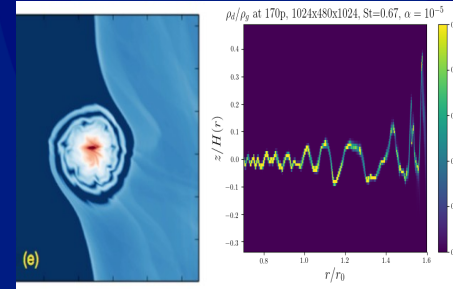
$10^{-7-8} R$

Streaming
Instability



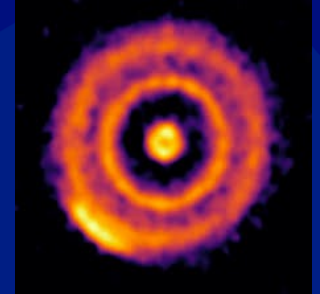
$0.01-0.1 H$

Meso-scale:
RWI, SGI, VSI,....



Several $H \sim 0.1 R$

Global disk



$\sim R$

(Small)



Also, radiation, B fields ...

(Large)

size

Two-Fluid 3D Dust-Gas LA-COMPASS code (and a new LANL/Artemis open-source code)

LA-COMPASS code: See Shengtai Li's talk (S. Li & H. Li)

1) Gas and multiple dust sizes:

$$\begin{aligned}\partial_t \rho_g + \nabla \cdot (\rho_g u_g) &= 0 \\ \partial_t u_g + u_g \cdot \nabla u_g &= \frac{-\nabla P + \tau_{vis} - \sum_i [\rho_{d,i} K_i (u_g - u_{d,i})]}{\rho_g} - \nabla \Phi \\ \partial_t \rho_{d,i} + \nabla \cdot (\rho_{d,i} u_{d,i}) &= \nabla \cdot \left((\rho_g + \rho_{d,i}) D \nabla \left(\frac{\rho_{d,i}}{\rho_{d,i} + \rho_g} \right) \right) \\ \partial_t u_{d,i} + u_{d,i} \cdot \nabla u_{d,i} &= K_i (u_g - u_{d,i}) - \nabla \Phi\end{aligned}$$

2) Dust size evolution: 151 size bins between 1 μm – 1 m:

(collaboration w. Birnstiel's group, Drążkowska+19, Y. Li+20, Laune+20)

3) Planet(s): planet(s) – disk interaction

Artemis:

**Dempsey et al. 2026,
in preparation**

**Drag force between gas and dust
species i (feedback term) Fu,
HL+14, Y. Li, HL'19**

- We have implemented some of these capabilities into Athena++ code (by Ho & S. Li; see also Huang & Bai; Pfeil papers)

#1: How will dust coagulation, bouncing affect the outcome w VSI

Ho, Li & Li, 2026, to be submitted

- VSI: Strong (isothermal) and weak w cooling time $\tau_c = \Omega_0^{-1}$
- no hydro viscosity
- using α_{coag} (10^{-3} above H, 10^{-5} midplane) for dust coagulation
- Disk radius: $r \in [16, 110] \text{ au}$; $R_0 = 40 \text{ au}$; $H/R = 0.08$; $\frac{z}{H} \in [-5, 5]$

$$\rho_g(R, z) = \rho_0 \left(\frac{R}{R_0} \right)^{-p} \exp \left(-\frac{z^2}{2H_g^2} \right)$$

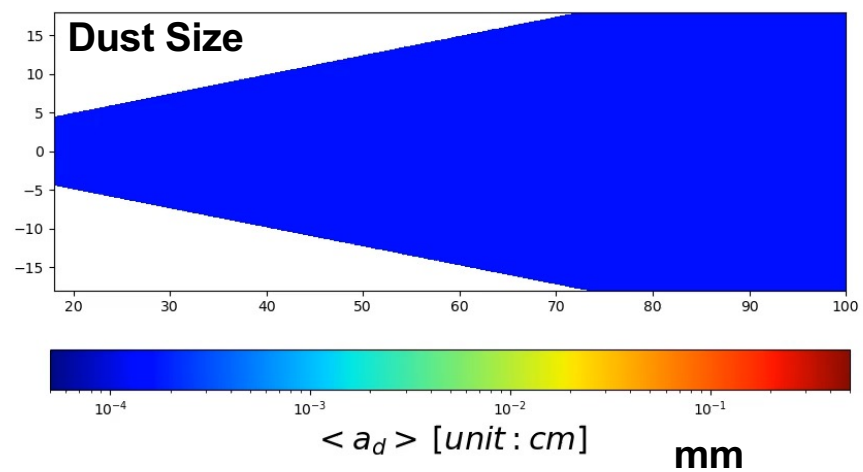
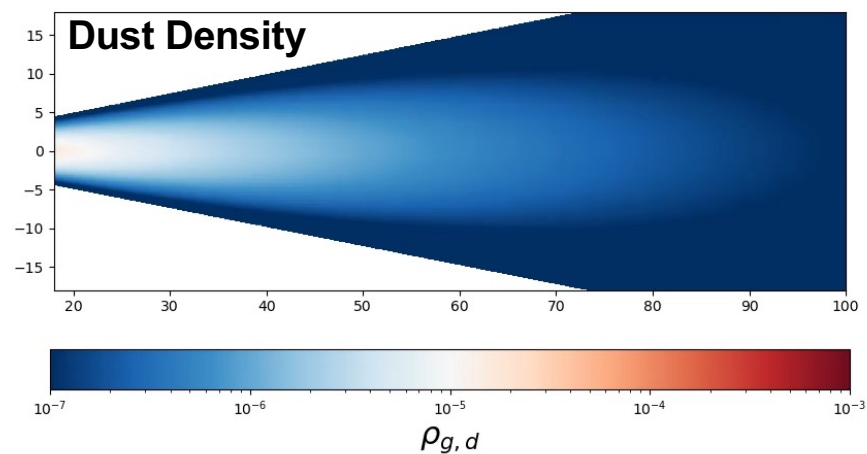
$$c_s^2(R) = c_s^2(R_0) \left(\frac{R}{R_0} \right)^{-q}$$

- Dust: initial D/G=0.0125 and initial size $1 \mu\text{m}$ (everywhere);
- Dust size range: $a_d \in [5 \times 10^{-5}, 5 \times 10^{-1}] \text{ cm}$, $v_f = 10 \text{ m/s}$
- Simulation: $t = 1614 \Omega_0^{-1}$ or ~ 160 orbits at 40 au.

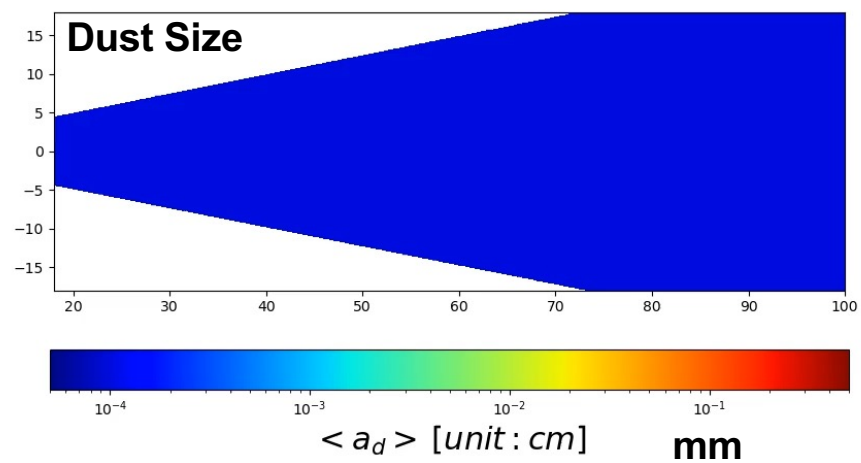
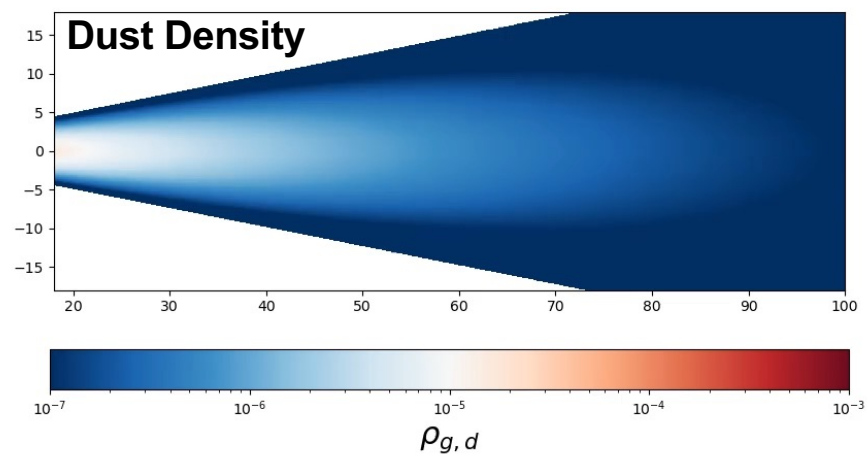
Bouncing

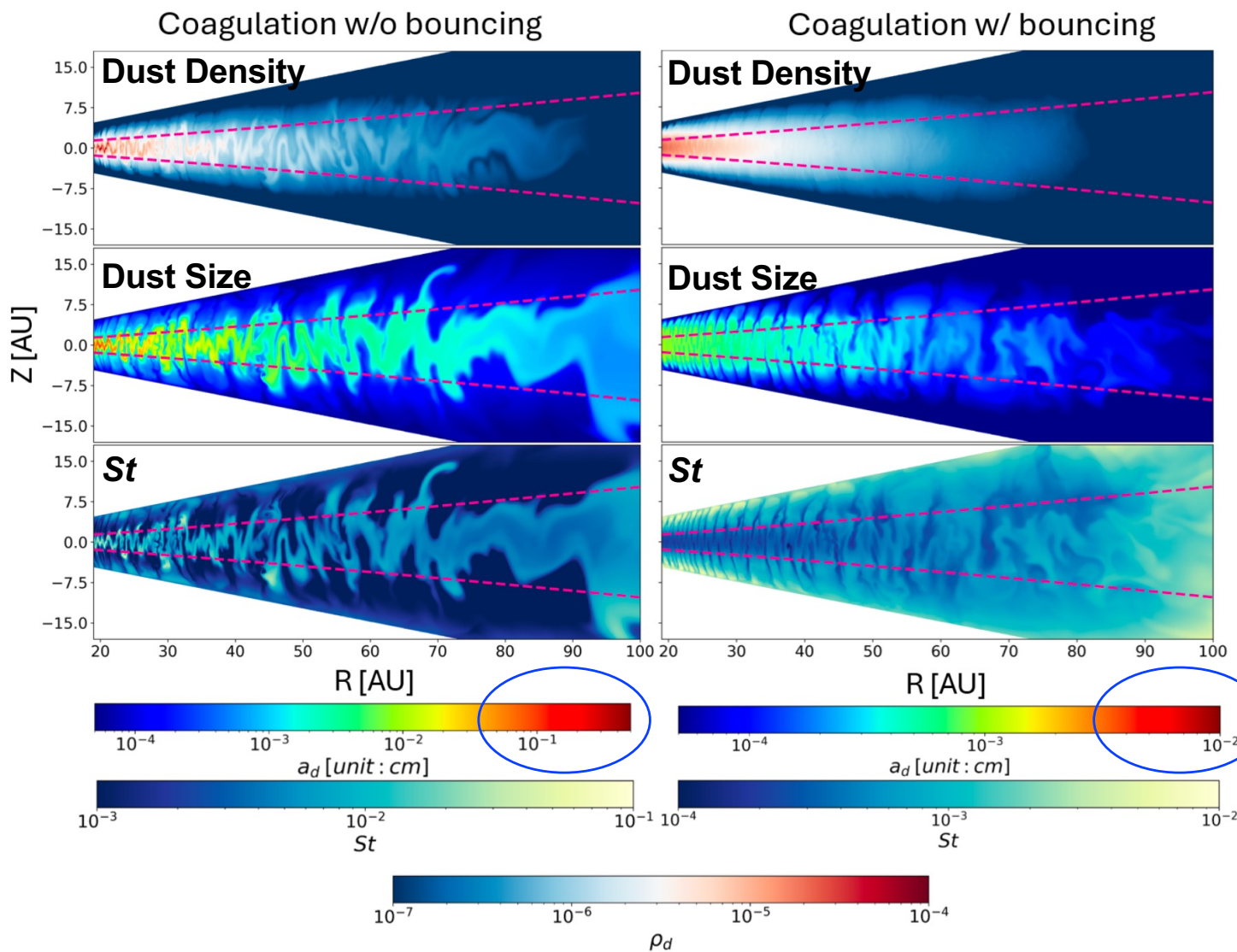
$$v_f = v_{f,0} \left(\frac{a}{a_{mono,0}} \right)^{-\frac{5}{6}}$$

Coagulation with Bouncing



Coagulation without Bouncing



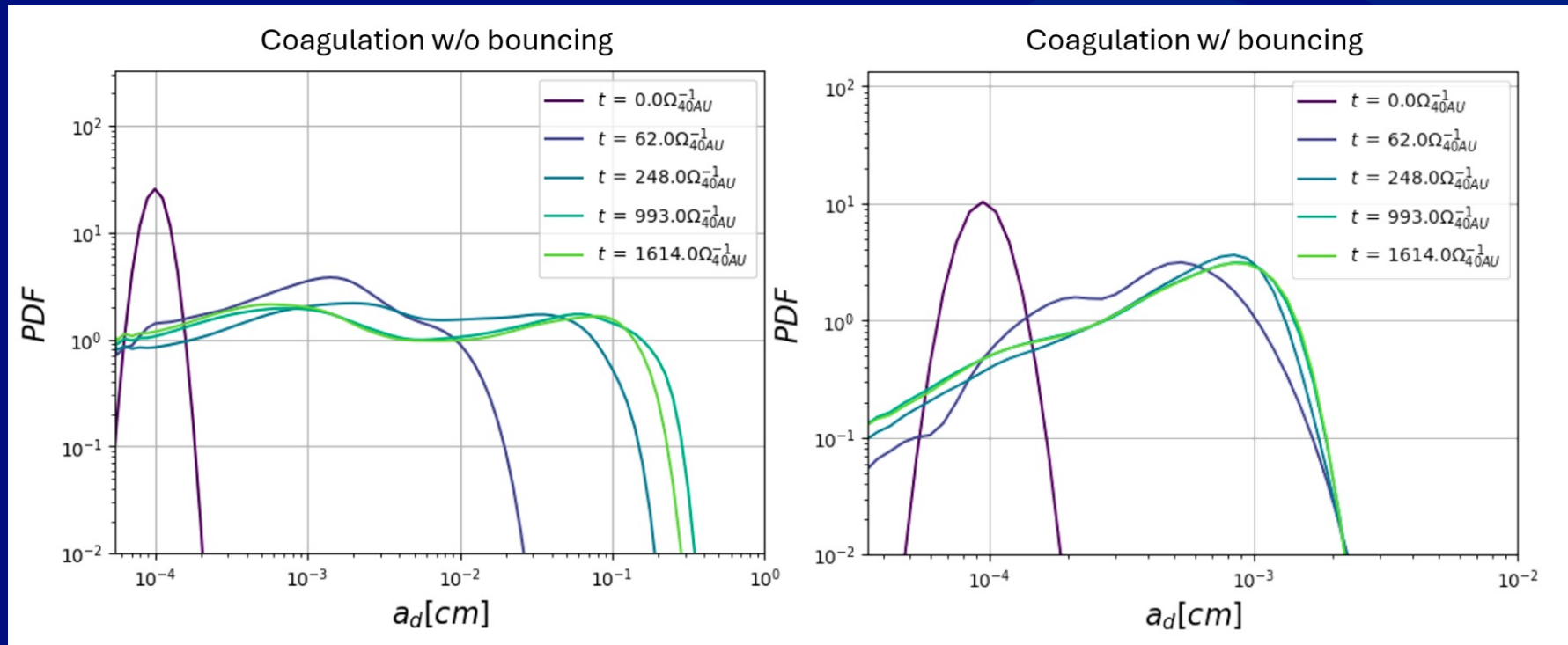


Bouncing has a strong effect

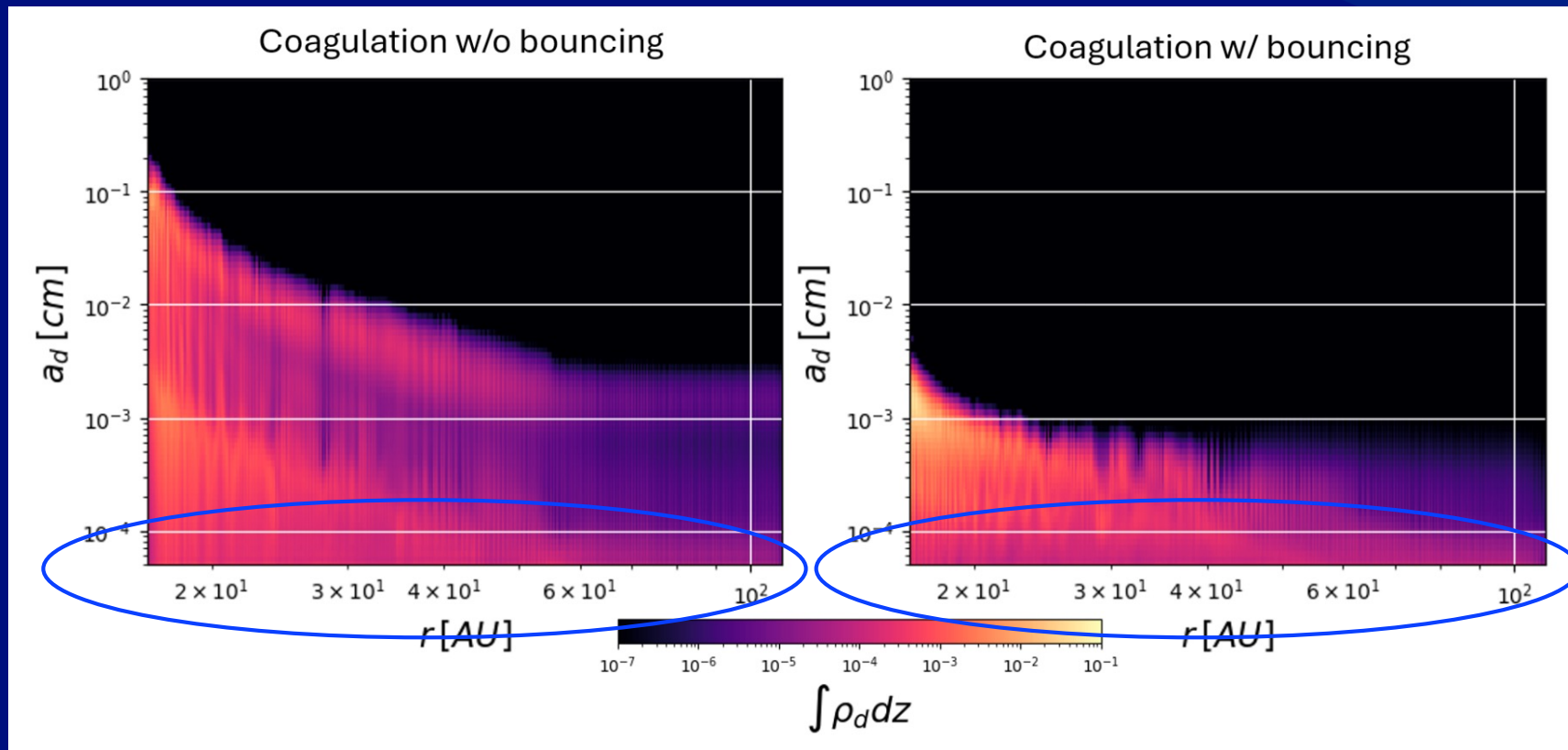
1. VSI can effectively maintains/stirs μm -size dust at higher latitudes;
2. Bouncing severely limits the dust size to $\sim 20 \mu m$;
3. Dust little settling in bouncing case;
4. W/o bouncing, dust size up to $\sim cm$.

Ho, Li & Li, 2026, to be submitted

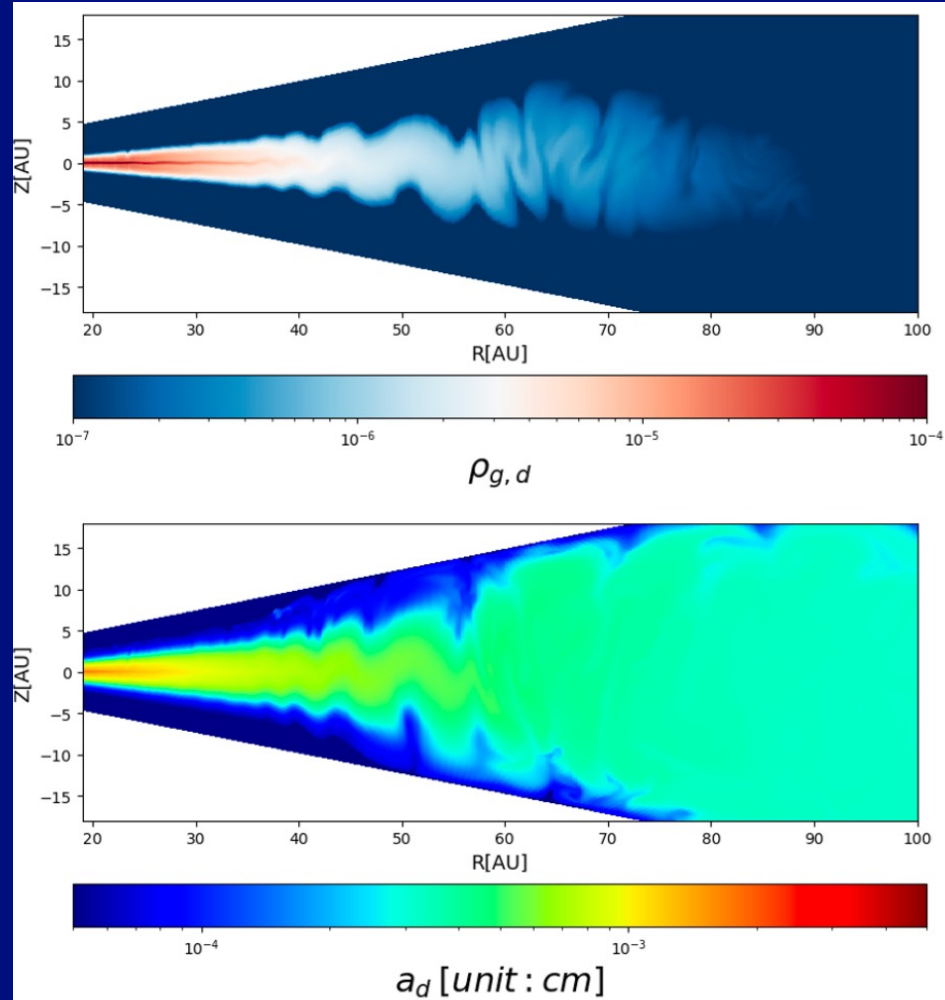
Size Evolution Very Different with Bouncing



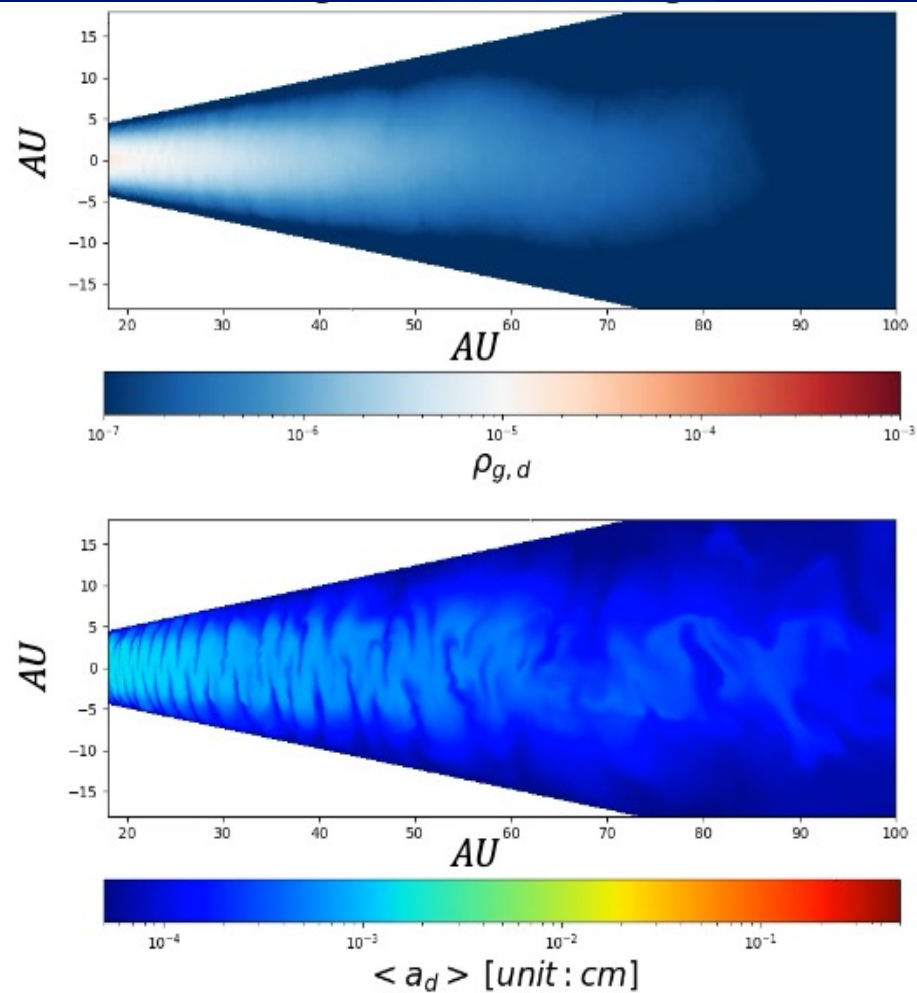
Still having a population of small dust



Weak VSI (w. bouncing)



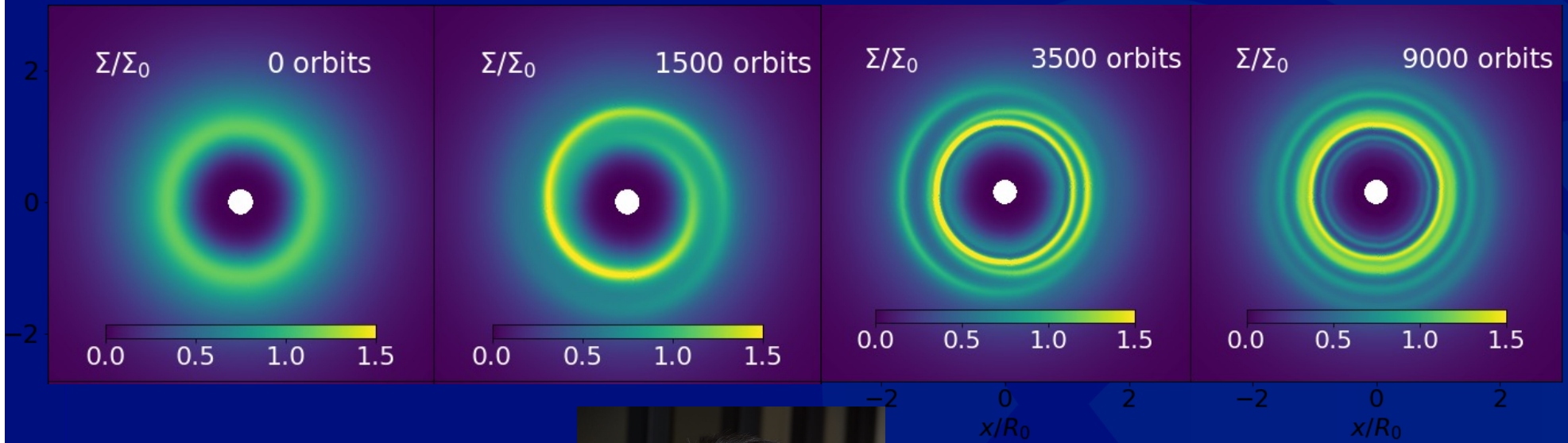
Strong VSI (w. bouncing)



Brief Summary and Next:

- Bouncing has a large impact on 2D dust size distribution, seem to help VSI to survive, and keep enough small dust at upper layers
- Need to include more sophisticated cooling treatment (See talks by Fukuhara and Pfeil, others)
- Radiative transfer (post-processing) for comparison with observations in different wavebands

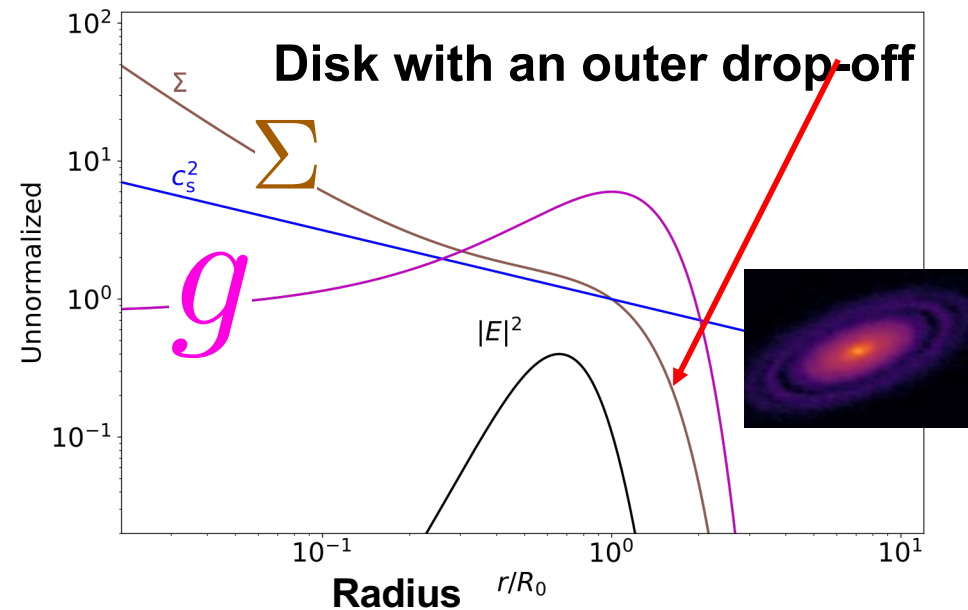
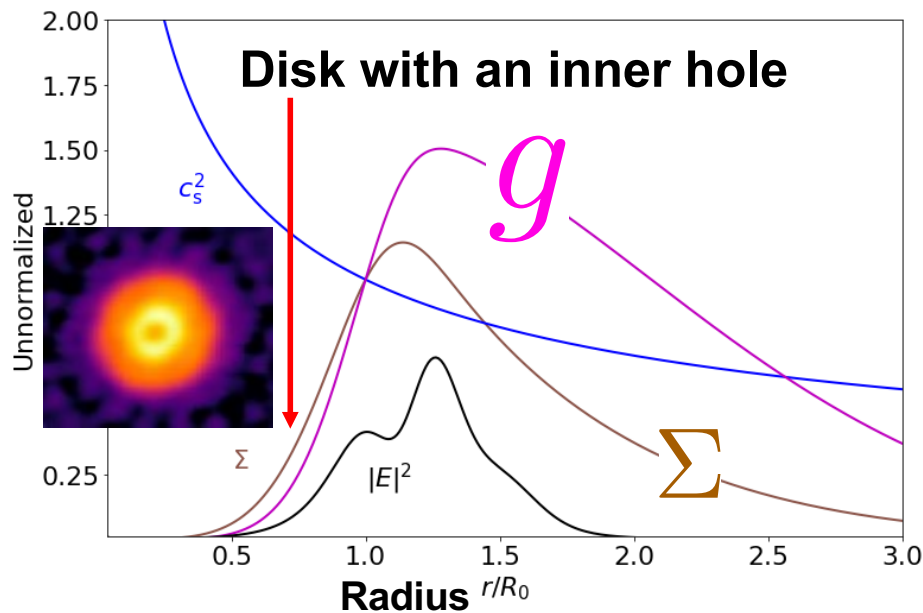
Example #2 : Eccentric Mode Instability in PPDs



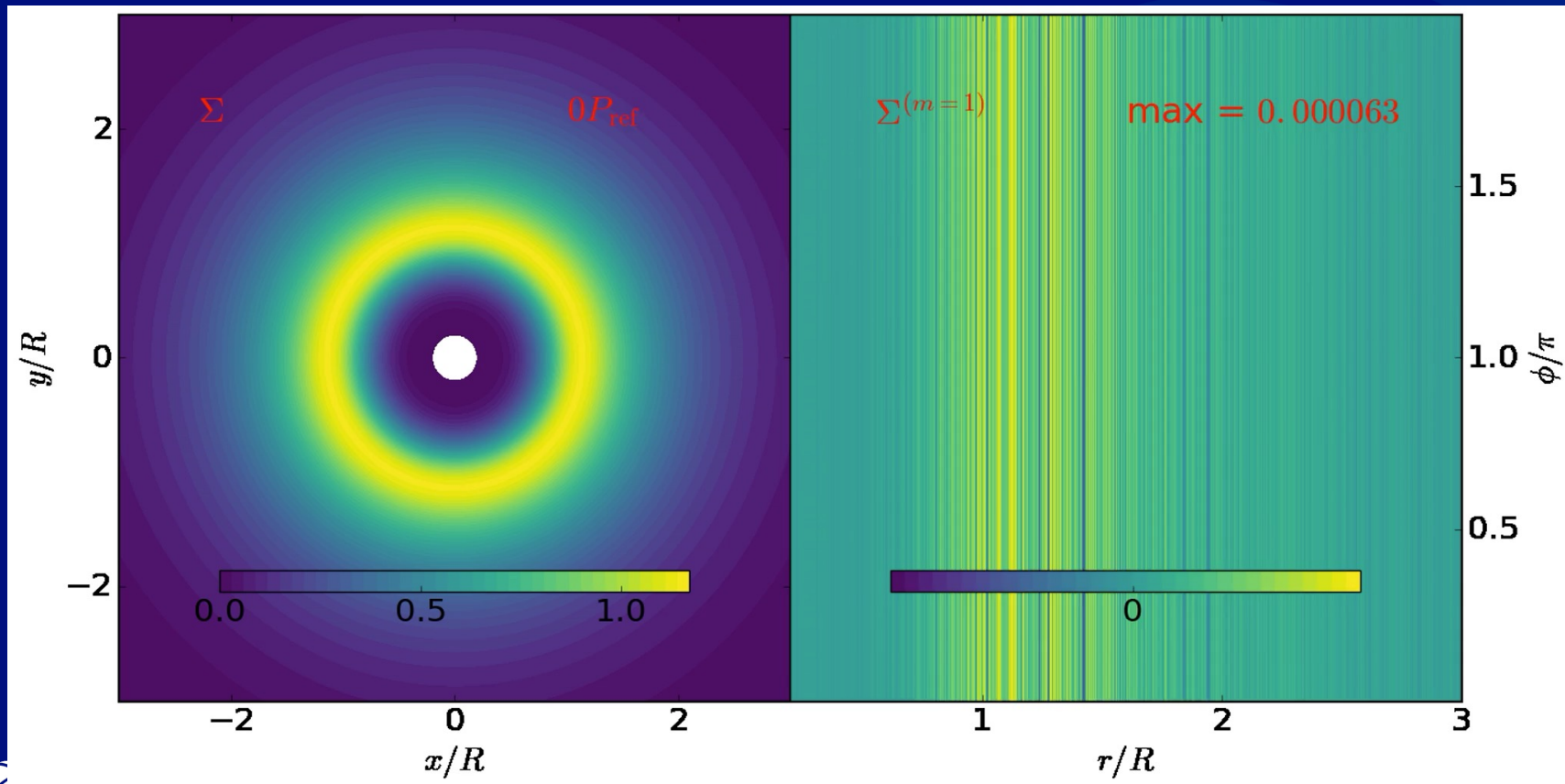
J. Li, A. Dempsey, HL, S. Li' 21

Self-gravitating Disk: Instability Profile and Criterion

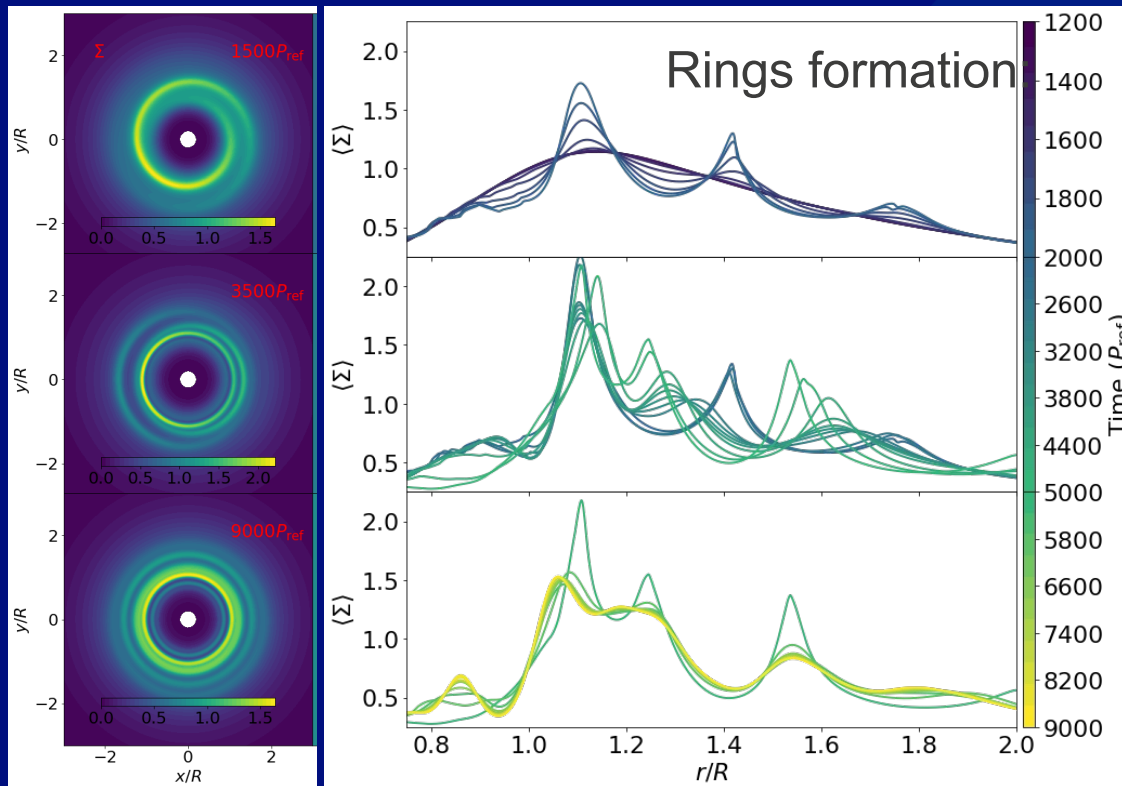
$$g(r) = \frac{\pi G \Sigma r}{c_{iso}^2} \quad Q(r) = \frac{c_s \Omega}{\pi G \Sigma} \quad g(r) * Q(r) = \frac{r}{h}$$



EMI: From Gas Spiral modes to Multiple Gas Rings

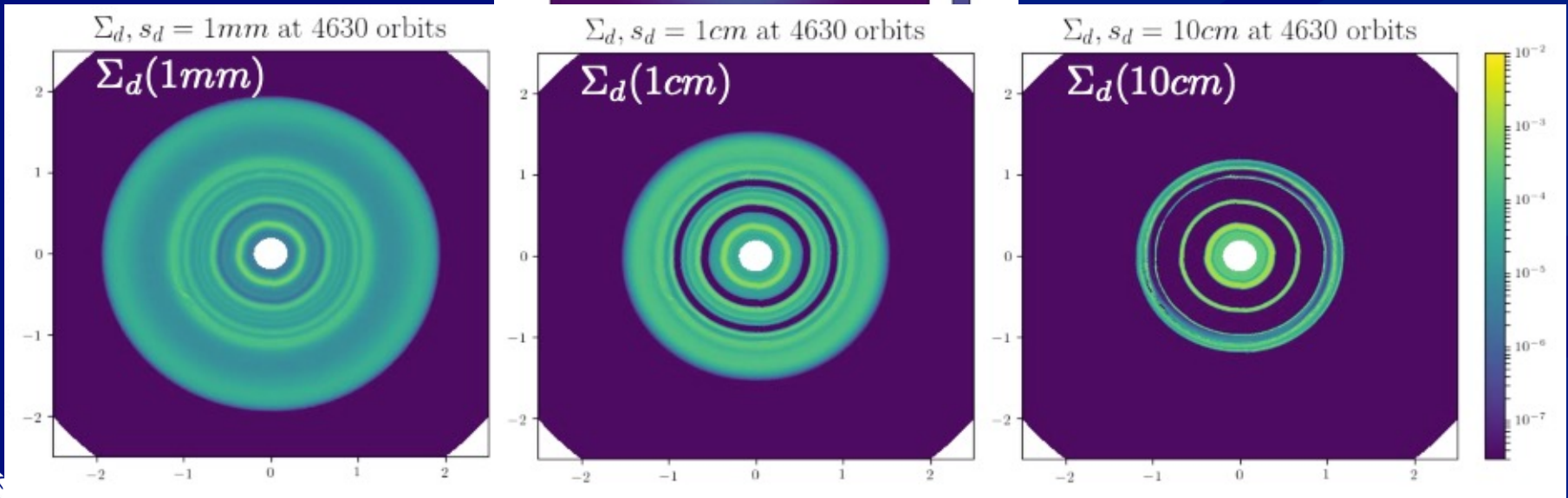
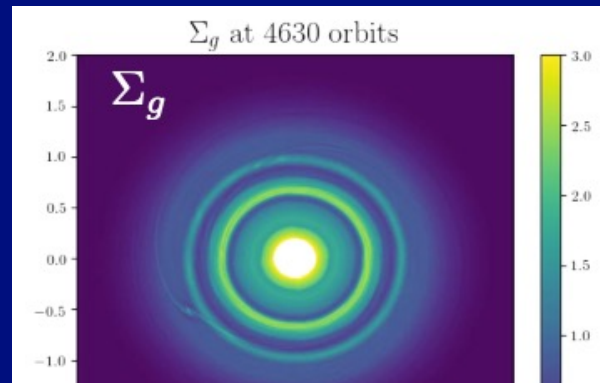


Mechanism: angular momentum exchange between s-mode and background disk flow



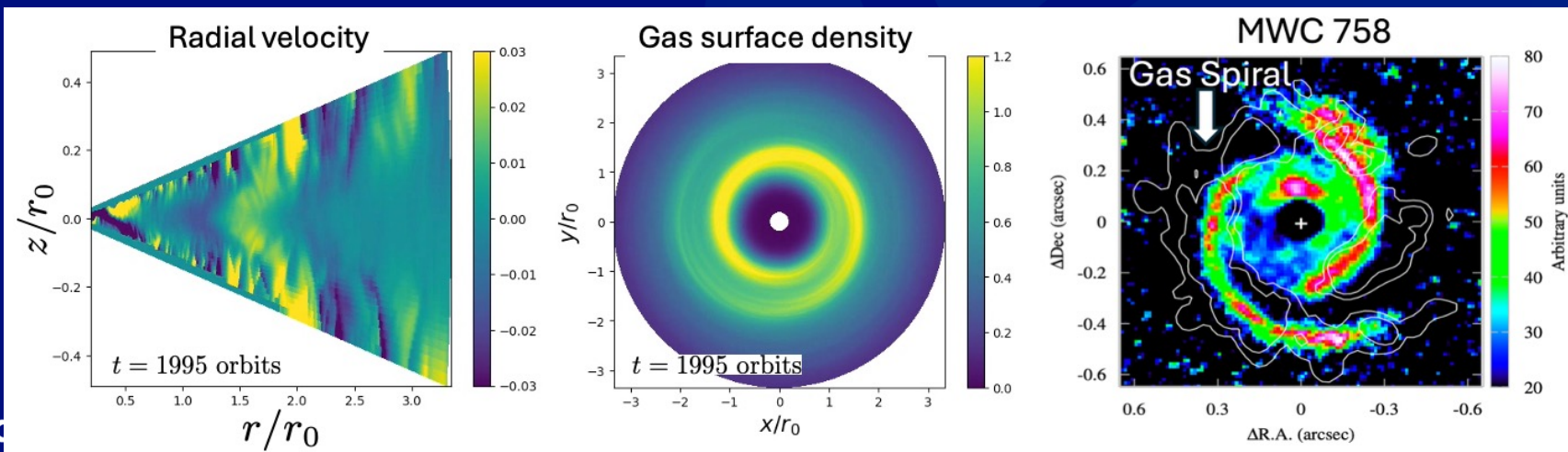
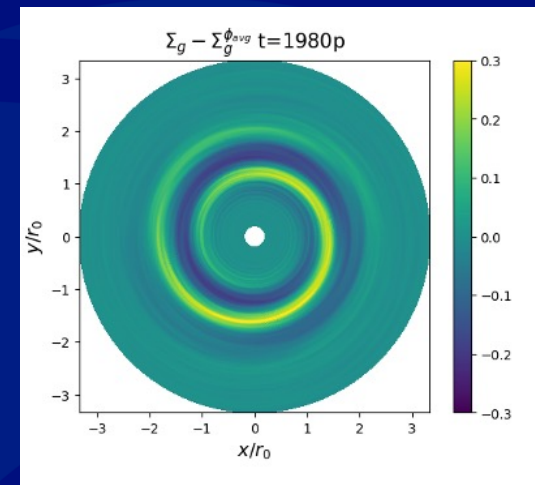
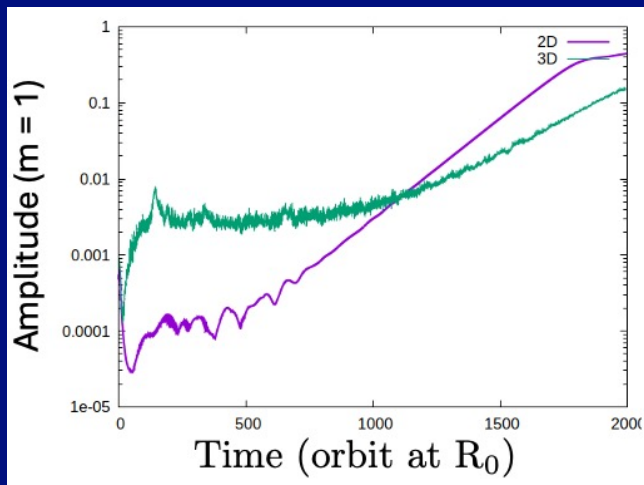
Different Dust Sizes w Coagulation

HL, S. Li, J. Li, in prep.

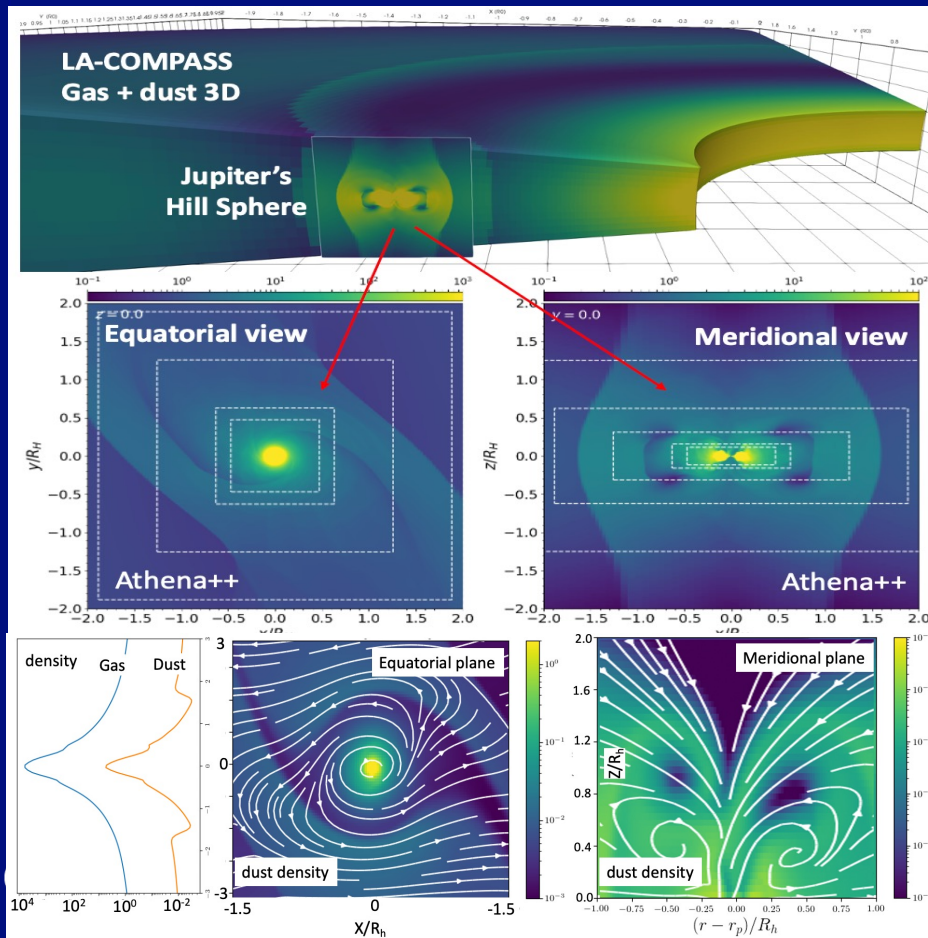


3D: EMI

HL, S. Li, J. Li, in prep



Example #3: Vortex Scenario for Planetesimal Formation with Dust Evolution in 3D Simulations



- Jupiter mass planet
- 3D gas + three dust sizes
- Low disk viscosity
- ~ 3500 rotations (0.57 Myr at 30 au)

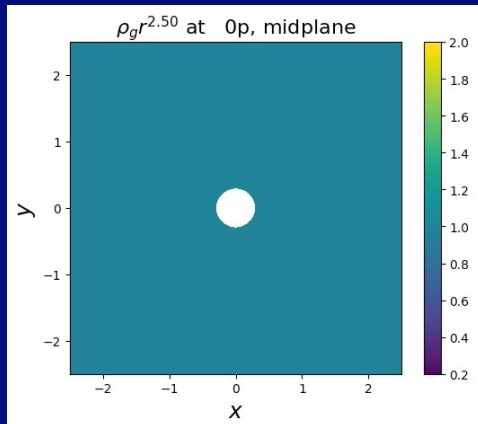
HL & S. Li 2025, in prep

Midplane Properties: vortex, ring and dust drifts

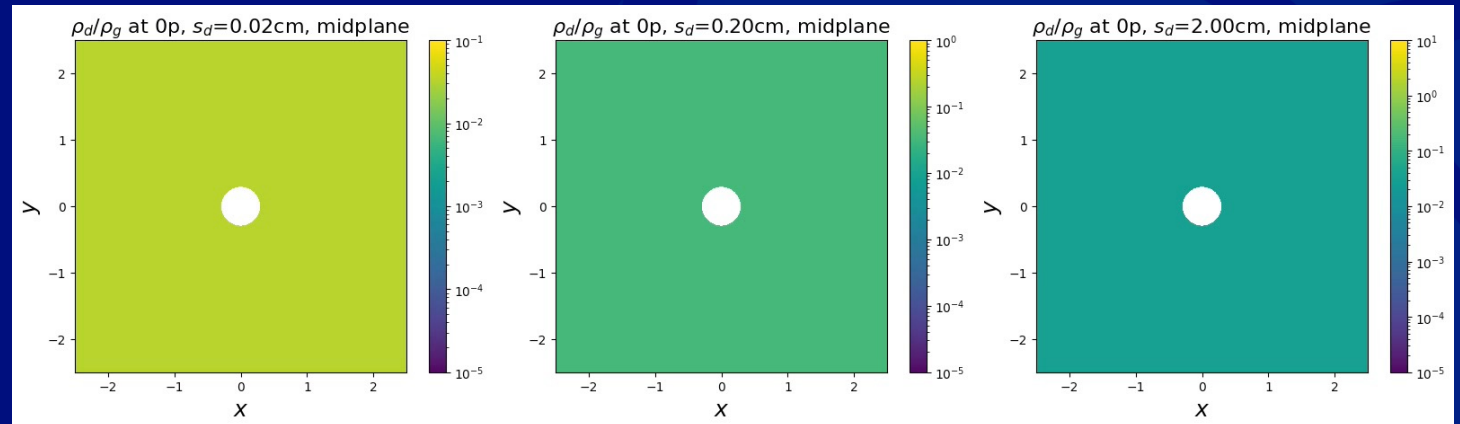
$$\rho_g * r^{2.5}$$

$$\rho_d / \rho_g$$

$$St = 2.6 \times 10^{-4}; 2.6 \times 10^{-3}; 2.6 \times 10^{-2};$$



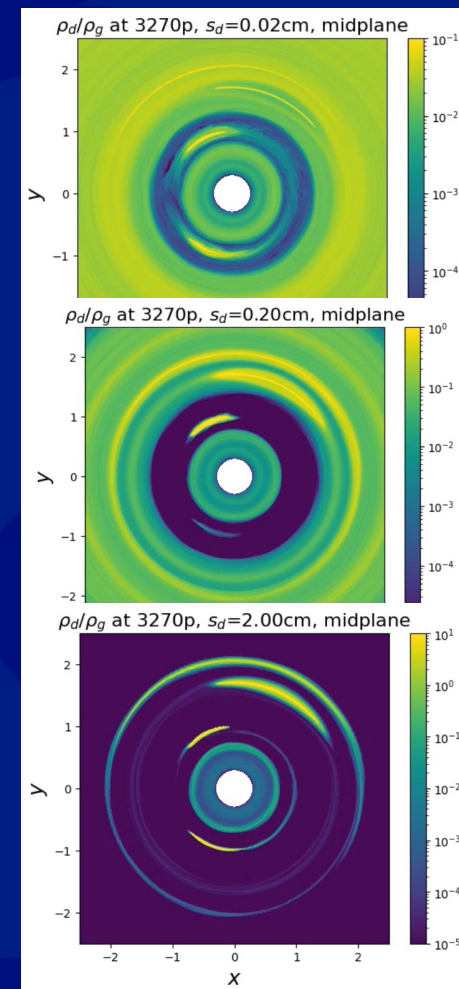
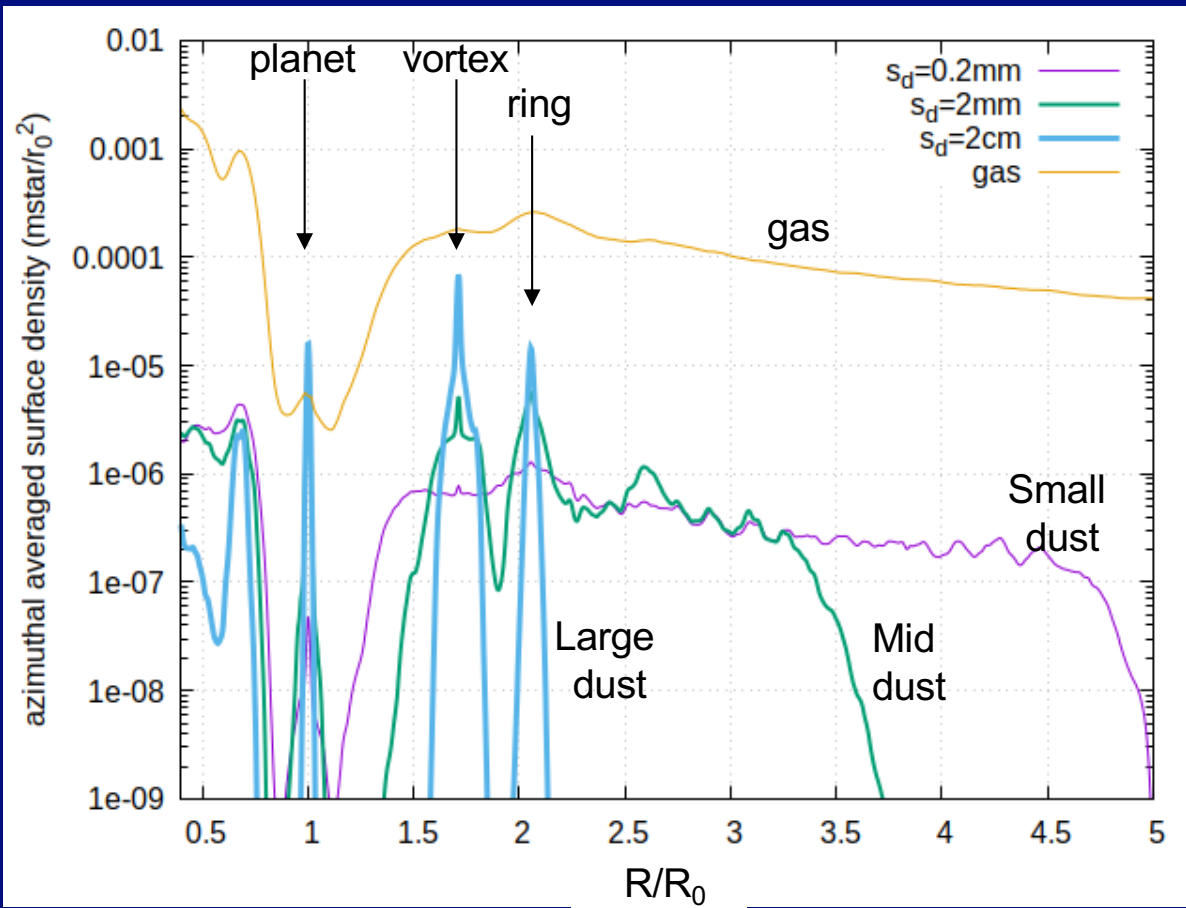
Gas density



Dust / Gas density ratio at midplane

HL & S. Li 2025, in prep

Dust Drift and Accumulation at Vortices, Rings and L4 and L5 Points of Planet



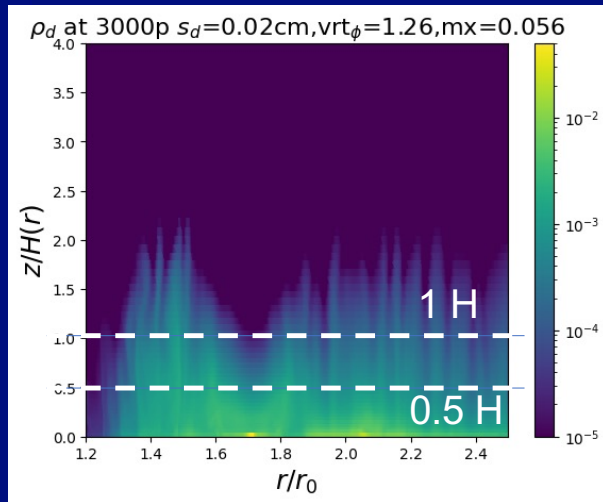
0.2 mm

2 mm

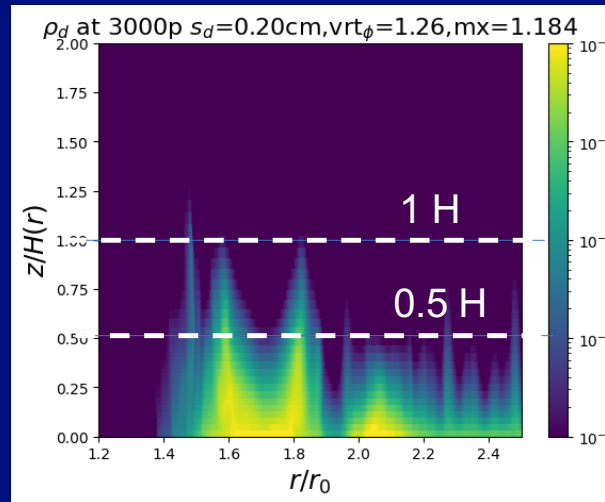
2 cm

Vertical Settling of different size dusts

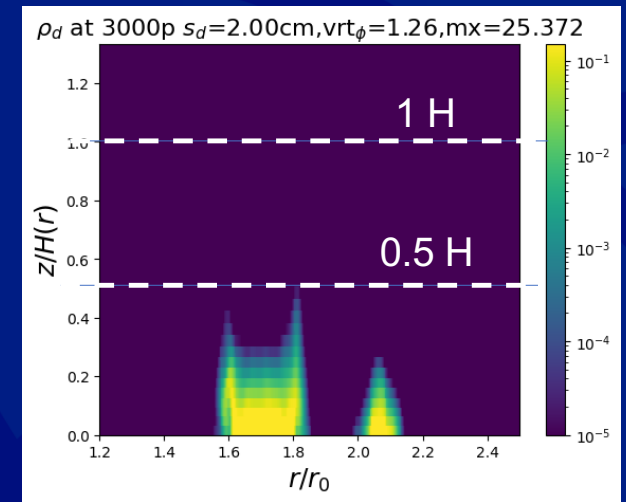
0.2 mm



2 mm



2 cm



$$St = 2.6 \times 10^{-4}; 2.6 \times 10^{-3}; 2.6 \times 10^{-2};$$

Summary and Next Steps:

- 1) Computational capabilities for including dust size evolution have been steadily improving, enabling comparison with multi-wavelength, gas and dust high-res observations.
- 2) Bouncing seems to help VSI survive and VSI stirs up small dust
- 3) Dust Coagulation can help SI
- 4) **Massive disks scenario very intriguing, disk instabilities, structures**

Next: better thermal dynamics (including radiation) will be important to determine various scenarios.